Solution

By the superposition principle, it may be observed that the final current distribution in the given circuit would be as shown in Fig. A.38.

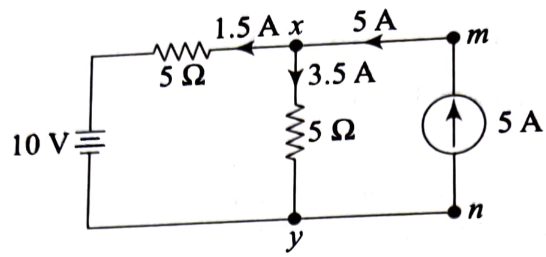

Fig. A.38

Hence, the voltage across 5 A current source may be calculated as

$$|V_{mn}| = |V_{xy}| = (5 \times 3.5) \quad \text{or} \quad [(5 \times 1.5) + 10] = 17.5 \text{ V} \qquad \text{(Ans)}$$

164. In a given circuit, while applying Thevenin's theorem, it is observed that Thevenin's equivalent voltage is 40 V and Thevenin's equivalent resistance is 3 Ω. If the load resistance for the same network is 5 Ω, give the details about the Norton's equivalent circuit diagram for the same circuit. [2009]

Solution

The Thevenin's equivalent circuit for the network as described in the problem is shown in Fig. A.39.

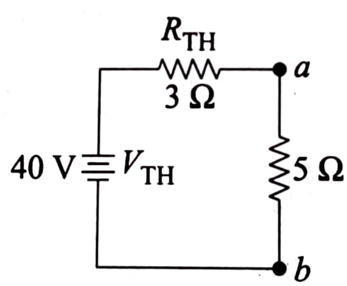

Fig. A.39

The equivalent Norton circuit for the same may be found to have a current source I_N across a Norton equivalent resistance R_N such that

$$I_N = \frac{V_{th}}{R_{th}} = \frac{40}{3} = 16.66 \text{ A} \quad \text{and} \quad R_N = R_{th} = 3\ \Omega \qquad \text{(Ans)}$$

The Norton's equivalent circuit for the above findings is shown in Fig. A.40.

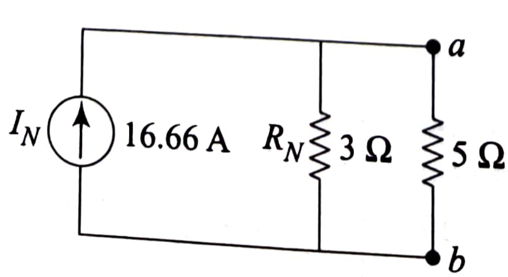

Fig. A.40

165. A circular iron ring wound with 500 turns of coil develops a magnetic flux of 20 milli-weber when the coil carries a current of 2 A. What would be the flux in the air gap if 1% of the mean length of the ring is removed? Assume that iron has a relative permeability of 1000.

Solution [2009]

The mmf for the given magnetic circuit is the ampere turn product, which gives 1000 AT for a current of 2 A and turns of 500. The reluctance of the complete iron ring may be calculated as the ratio of mmf and magnetic flux, which gives

$$\Re = \frac{\text{mmf}}{\varnothing} = \frac{1000}{20 \times 10^{-2}} = 5.0 \times 10^4 \, \frac{\text{AT}}{\text{Wb}}$$

When 1% of the mean length is removed, the new reluctance becomes the series combination of two reluctances comprising 99% of the mean length as iron and 1% of the mean length as air. Hence, the new reluctance is found as

$$\Re' = \Re_l + \Re_a = [0.99(5.0 \times 10^4)] + \left[\frac{0.01(5.0 \times 10^4)}{1000}\right]$$

$$= 4.95 \times 10^4 \, \frac{\text{AT}}{\text{Wb}}$$

Hence, the new flux in the air gap would be

$$\phi' = \frac{\text{mmf}}{\Re'} = \frac{1000}{4.95 \times 10^4} = 20.2 \times 10^{-3} \text{ Wb} = 20.2 \text{ milli-weber}$$

(Ans)

166. Two impedances of values $(5 + j6)\ \Omega$ and $(8 - j3)\ \Omega$ are connected in parallel. What would be the net equivalent impedance of the combination? What would be the phase of the current with respect to the supply voltage when the combination is excited from a 200 V, 50 Hz single-phase AC supply? [2009]

Solution

In a parallel combination, the net equivalent impedance may be calculated as

$$Z_{eq} = \frac{Z_1 Z_2}{Z_1 + Z_2} = \frac{(5 + j6)(8 - j3)}{(5 + j6) + (8 - j3)} = (4.8 + j1.43)\ \Omega \qquad \text{(Ans)}$$

Hence, the current becomes

$$I = \frac{V}{Z_{eq}} = \left(\frac{200 + j0}{4.8 + j1.43}\right) = (40\angle -16.64°) \text{ A}$$

Thus, it is found that the phase of the current with respect to the voltage is $\varphi = 16.64°$ (lag) (Ans)

167. Three identical impedances connected in star fashion draw a line current of $(5\angle -30°)$ A when connected across a 400 V, 50 Hz, three-phase AC supply. Find the resistance and reactance of impedance per phase. [2009]

Solution

Given that $V_L = 400\angle 0°$ V and $I_L = 5\angle -30°$ A

Since the network is star connected,

$$V_{ph} = \frac{V_L}{\sqrt{3}} = 231\angle 0° \text{ V}$$

and

$$I_{ph} = I_L = 5\angle -30° \text{ A}$$

So, impedance per phase may be calculated as,

$$Z_{ph} = \frac{V_{ph}}{I_{ph}} = (46.2\angle 30°)\ \Omega = (40 + j23)\ \Omega$$

Knowing that $Z_{ph} = R_{ph} + jX_{ph}$, resistance and reactance per phase may be expressed as, $R_{ph} = 40\ \Omega$ and $X_{ph} = 23\ \Omega$ (inductive). (Ans)

168. A single-phase transformer develops 200 V at the secondary terminals on no-load condition. If the secondary winding has 1000 turns, find the maximum flux in the core. Assume a 20 V, 50 Hz single-phase AC supply at the primary. [2009]

Solution

The secondary induced emf is

$$E_2 = 4.44 f \phi_m T_2$$

Hence, the maximum flux in the core becomes

$$\phi_m = \frac{E_2}{4.44 f T_2} = \frac{200}{4.44 \times 50 \times 1000} = 90 \text{ milli-weber} \qquad \text{(Ans)}$$

169. A DC shunt generator develops 200 V on no load, running at 1200 rpm. If it has four poles and 100 lap wound armature conductors, calculate the flux per pole. Also calculate the shunt field current if the resistance of the shunt field is 200 Ω. [2009]

Solution

From the emf induced expression for a DC shunt generator, we may find the expression for the flux by taking $A = P = 4$ for lap winding. So,

$$\phi = \frac{60.A.E_0}{P.Z.N} = \frac{60 \times 200}{100.1200} = 0.1 \text{ Wb} \qquad \text{(Ans)}$$

Also, the shunt field current is

$$I_{sh} = \frac{V}{R_{sh}} = \frac{200}{200} = 1\text{ A}$$

(Ans)

170. A three-phase, four-pole, induction motor runs at 1% slip when supplied from a 400 V, 50 Hz three-phase AC supply. Calculate the rotational speed of the revolving magnetic field and running speed of the motor. [2009]

Solution

From the given data, the rotational speed of the revolving magnetic field is same as the synchronous speed, which may be found as

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500\text{ rpm}$$

Since the slip is 1% (i.e. $s = 0.01$), the running speed of the motor may be calculated with the help of the slip expression

$$s = \frac{N_s - N_r}{N_s}$$

This gives $N_r = 1485$ rpm (Ans)

171. Applying the nodal analysis to the circuit given in Fig. A.41, calculate the voltages at all possible nodes. [2009]

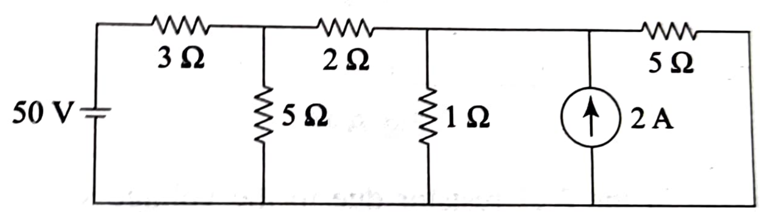

Fig. A.41

Solution

The above circuit may be reduced to the circuit shown in Fig. A.42 with the help of source transformation.

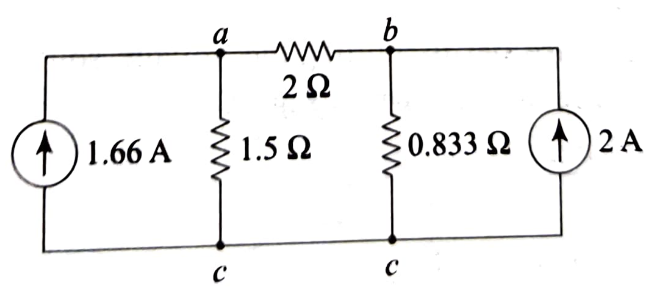

Fig. A.42

The KCL expressions for the nodes a and b may be written by assuming that the voltages at the nodes a, b, and c are V_a, V_b, and V_c respectively. Further, it may be assumed that $V_a > V_b > V_c$ and $V_c = 0$. These expressions become

$$\frac{V_a - V_b}{2} + \frac{V_a}{1.5} = 1.66 \text{ and } \frac{V_a - V_b}{2} + 2 = \frac{V_b}{0.833}$$

On solving these two expressions, it is found that $V_a = 2.21$ V and $V_b = 1.85$ V (Ans)

172. **Applying the superposition principle to the circuit given in Fig. A.43, calculate the current through the 2 Ω resistor.** [2009]

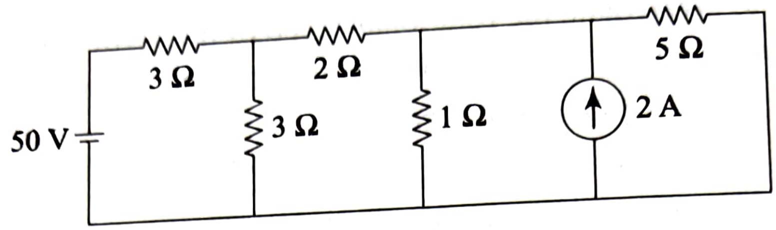

Fig. A.43

Solution

In the first case, let us nullify the current source by opening it and represent the circuit as shown in Fig. A.44.

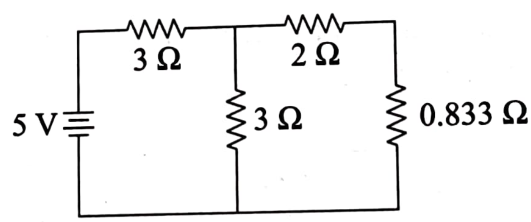

Fig. A.44

The current in the 2 Ω resistor due to the voltage source alone may be found with the help of the circuit as per the expression given below. This current is, say,

$$I_1 = \left[\frac{5}{3} + \left(\frac{3 \times 2.833}{5.833}\right)\right]\left[\frac{3}{5.833}\right] = 0.58 \text{ A}$$

In the second case, let us nullify the voltage source by shorting it and represent the circuit as shown in Fig. A.45.

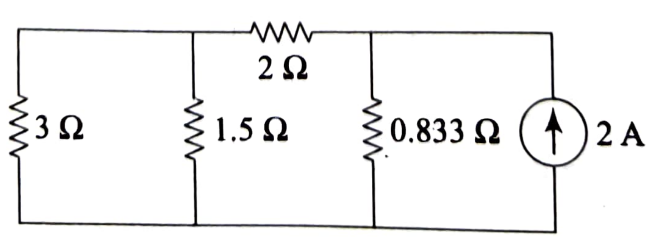

Fig. A.45

The current in the 2 Ω resistor due to the current source alone may be found with the help of the circuit as per the expression given below. This current is, say,

$$I_2 = \left[\frac{0.833}{0.833+2} + \left(\frac{3\times 1.5}{4.5}\right)\right]^{[2]} = 0.41 \text{ A}$$

Hence, the net current in the 2 Ω resistor may be found as

$$I = I_1 - I_2 = 0.17 \text{ A}$$

(Ans)

173. A rectangular shaped iron core is prepared out of cylindrical iron rods of different cross section and permeability, as shown in Fig. A.46. [2009]

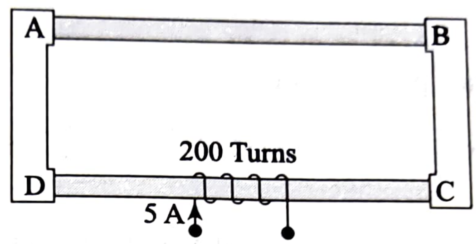

Fig. A.46

Calculate

(i) reluctance of individual section,
(ii) total reluctance of the core,
(iii) flux in the core, and
(iv) flux density in individual section.

Details of these parameters are given in the table below.

Section	*Length* (l)	*Cross-sectional area* (A)	*Relative permeability* (μ_r)
AB	20 cm	5 cm^2	1000
BC	10 cm	2 cm^2	1100
CD	20 cm	3 cm^2	1200
DA	10 cm	4 cm^2	1500

Solution

The following hints may be used to solve this problem.

(i) Reluctance for any section-x may be found by using the formula

$$\Re_x = \frac{l}{\mu_0 \mu_r A}.$$

(Ans)

(ii) Total reluctance of the core may be found by summing the individual reluctance values. (Ans)

(iii) Flux in the core may be found by taking the ratio of mmf (i.e. the ampere turn product) and the total reluctance $\varnothing = \dfrac{\text{mmf}}{\Re}$. (Ans)

(iv) Flux density in any section-x may be found by using the formula

$$B_x = \phi \frac{1}{A}.$$ (Ans)

174. A resistance of 10 Ω, an inductor of inductance of 2 H, and a capacitor of capacitance 100 microfarad are connected to a single-phase 230 V AC source of varied frequency. Find the (i) current, (ii) power factor, and (iii) active power consumption corresponding to the supply frequencies of 50 Hz and 100 Hz, respectively. [2009]

Solution

Given values for this problem are

$$R = 10\ \Omega,\ L = 2\ \text{H},\ C = 100 \times 10^{-6}\ \text{F},\ V = 230\ \text{V},$$

and $f = 50$ Hz

(a) In the first case, let us consider 50 Hz frequency for the supply. The reactances may be found as per the formulae

$$X_L = 2\pi fL \text{ and } X_C = \frac{1}{2\pi fC}$$

Then, the impedance may be found as per the formula $Z = R + j(X_L - X_C)$ and the current may be found as $I = \frac{V}{Z}$. Further, the power factor may be found as the ratio of the resistance to the impedance magnitude, i.e. $\cos\varphi = \frac{R}{|Z|}$. The active power consumption may be found as $P = V.I.\cos\varphi$.

(b) In the first case, let us consider 100 Hz frequency for the supply. The required unknowns may be found in a similar way as done in the first case.

175. A capacitor is charged from a DC supply of 100 V through a resistor in series having a resistance of 100 Ω. If the time constant for the given set up is 15 milli-seconds, calculate the value of the capacitance. Also calculate the time requirement for the capacitor to acquire 90% of the steady state charge assuming zero initial charge in the capacitor. [2009]

Solution

Time constant for a series RC circuit is expressed as $\tau = RC$. Therefore, the value of the capacitance may be

$$C = \frac{\tau}{R} = \frac{15 \times 10^{-2}}{100} = 1.5 \text{ microfarads}$$ (Ans)

In the second part, the time may be calculated from the charge expression

$$q = Q\left(1 - e^{\frac{\tau}{RC}}\right)$$

where $q = 0.9Q$ corresponding to 90% of the steady state charge. (Ans)

176. Calculate the combined impedance and admittance in polar form when three impedances of values $(5\angle 30°)\ \Omega$, $(3 + j6)\ \Omega$, and $(4 - j8)\ \Omega$ are connected in series. [2009]

Solution

The three given impedances may be expressed in rectangular form as

$$Z_1 = (5 \angle 30°)\ \Omega = (4.33 + j2.5)\ \Omega,\ Z_2 = (3 + j6)\ \Omega,$$

and $$Z_3 = (4 - j8)\ \Omega$$

When connected in series, the combined equivalent impedance becomes $11.34 \angle 0.4°\ \Omega$. (Ans)

Hence, the equivalent admittance becomes $Y = 1/Z = 0.088 \angle -2.5°$ mho. (Ans)

177. Find out the expected speed of revolution of the steam turbine if the turbo generator to be installed in a thermal power plant has a two-pole structure and the system frequency is to be maintained at 50 Hz. [2009]

IMP

Solution

The expected speed for a turbo generator is the synchronous speed given by the expression

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{2} = 3000 \text{ rpm} \quad \text{(Ans)}$$

178. A PMMC measuring instrument gives full-scale deflection when connected across 100 V DC supply by taking a current of 5 A. What should be the value of a suitable shunt resistance in order to enable the same instrument to read up to 50 A. Also find out the voltage that may be measured safely by the same instrument when a series resistance of 50 Ω is connected in series with the same instrument. [2009]

Solution

From the given data, the resistance of the meter is

$$R = \frac{100}{5} = 20\ \Omega$$

As per the current division principle, the shunt resistance (R_{sh}) should be such that the meter having a safe current capacity of 5 A may be used as an ammeter to read up to 50 A. This expression is given by

$$5 = 50\left(\frac{R_{sh}}{R_{sh} + R}\right)$$